

PROUD ADDITION
The launch ceremony of the Kalvari-class submarine INS *Vagsheer* in Mumbai, Apr. 20, 2022

TKMS has tied up with the Mumbai-based Mazagon Dock Shipbuilders Ltd (MDL), and Indian firm Larsen & Toubro (L&T) with Navantia to build the submarines. The Germans are said to have gained the upper hand in the FETs with their Type 212 submarines, as Navantia failed the navy's stringent AIP requirements. The navy has submitted its report to the defence ministry.

P75(I) assumes importance because India's submarine fleet is down to just 16 in contrast to the fast increasing fleets of its adversaries, China and Pakistan. Only half the Indian subs are battle-ready at any given time; the rest are undergoing upgrades. True, with the commissioning of INS *Arighat* on August 29, India has two indigenously made nuclear-powered ballistic missile submarines (SSBNs), has a plan to acquire a fleet of nuclear-powered conventional attack submarines (SSNs) and already operates five diesel-electric attack submarines of the Scorpene-class (that have the same base blueprint and are named after the first vessel of the type). But it needs to beef up its ageing fleet of conventional submarines, a gap the latest vessels would close.

Though India's submarine force began its journey in 1967 with the induction of the Soviet-origin Kalvari (S-23), and the navy's submarine strength went up to 21 diesel-powered attack subs by the mid-1980s, submarine programmes have hit rough weather in the last four decades. They have been plagued by delays due to cost overruns and technological challenges, and acquisitions have attracted a volley of kickback accusations, be it over the purchase of submarines from the German firm Howaldtswerke-Deutsche Werft (HDW) in 1987, or the Scorpene deal in 2005.

THREAT FROM CHINA, PAKISTAN

China's People's Liberation Army Navy (PLAN) has rapidly expanded and modernised its now 50-strong submarine fleet, with both conventional diesel-electric submarines and nuclear-powered vessels that include the Yuan-class conventional submarines, the Shang-class SSNs and the Jin-class SSBNs, which are capable of launching JL-2 missiles with intercontinental reach. This growing heft underscores